

Phase 3 Study of Xanomeline-Trospium

Protocol Number: CT-2024-001

Sponsor: NeuroPharma Therapeutics

1. Background and Rationale

Alzheimer's disease is a progressive neurodegenerative disorder affecting approximately 6.5 million adults in the United States. Current symptomatic treatments, including acetylcholinesterase inhibitors, provide modest benefit but are associated with significant gastrointestinal side effects. Xanomeline is a muscarinic acetylcholine receptor agonist that has demonstrated cognitive benefits in prior clinical trials, but its development was limited by cholinergic side effects. Trospium is a peripherally-restricted muscarinic antagonist that does not cross the blood-brain barrier, which may mitigate peripheral cholinergic adverse events when co-administered with xanomeline.

2. Study Objectives

Primary Objective: To evaluate the efficacy of xanomeline-trospium combination therapy compared to placebo on cognitive function in patients with mild-to-moderate Alzheimer's disease, as measured by change from baseline on the Alzheimer's Disease Assessment Scale-Cognitive Subscale (ADAS-Cog) at Week 24.

Secondary Objectives: (1) To evaluate the effect of xanomeline-trospium on global clinical status as measured by the Clinician's Interview-Based Impression of Change Plus Caregiver Input (CIBIC+); (2) To evaluate the effect on activities of daily living using the Alzheimer's Disease Cooperative Study - Activities of Daily Living Inventory (ADCS-ADL); (3) To assess the safety and tolerability profile.

3. Study Design

This is a multicenter, randomized, double-blind, placebo-controlled, parallel-group Phase 3 study. Approximately 800 patients will be randomized in a 1:1 ratio to receive either xanomeline-trospium combination therapy or matching placebo for 24 weeks of double-blind treatment, followed by a 28-week open-label extension period. Randomization will be stratified by disease severity (mild vs. moderate) and concomitant use of cholinesterase inhibitors (yes vs. no).

4. Inclusion Criteria

1. Male or female patients aged 50-85 years inclusive. 2. Diagnosis of probable Alzheimer's disease per NIA-AA criteria. 3. Mini-Mental State Examination (MMSE) score of 14-26 inclusive at screening. 4. Rosen Modified Hachinski Ischemic Score ≤ 4 . 5. Stable doses of cholinesterase inhibitors for at least 3 months prior to screening, if applicable. 6. Study partner available who has at least 10 hours per week of contact with the patient. 7. Written informed consent obtained from patient or legally authorized representative.

5. Exclusion Criteria

1. Current diagnosis of any other form of dementia. 2. Evidence of clinically relevant cerebrovascular disease. 3. Major depressive episode within the past 6 months. 4. History of alcohol or substance abuse within the past 2 years. 5. Clinically significant hepatic or renal impairment. 6. Use of anticholinergic medications with significant central nervous system activity. 7. History of seizures or epilepsy. 8. Participation in any other interventional clinical trial within 30 days prior to screening.

6. Statistical Considerations

The primary efficacy analysis will be performed using a mixed model for repeated measures (MMRM) on the change from baseline in ADAS-Cog score at Week 24. The analysis will include treatment group, visit, stratification factors, and treatment-by-visit interaction as fixed effects, with baseline ADAS-Cog score as a covariate. A sample size of 800 patients (400 per arm) provides approximately 90% power to detect a treatment difference of 2.5 points on ADAS-Cog at Week 24, assuming a standard deviation of 8 points and a two-sided alpha level of 0.05.